Paper 23 - FoldForge Protein Folding

CQE-paper-23
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Construction Basis

Apply the same ribbon/bifurcation logic to protein-chain fold hypotheses with contact-map and topology receipts.

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- production paper body pending rewrite: production\papers\CQE-paper-23\PAPER-BODY.md
- paper kernel manifest: production\paper-kernels\papers\CQE-paper-23\paper_kernel_manifest.json
- body: production\papers\CQE-paper-23\PAPER-BODY.md
- source: production\papers\CQE-paper-23\SOURCE.md
- formal: production\papers\CQE-paper-23\01-CQE-formal\FORMAL.md
- tool: production\papers\CQE-paper-23\02-CQE-tool\TOOL.md
- workbook: production\papers\CQE-paper-23\03-CQE-workbook\WORKBOOK.md
- recovered_output: production\lib-forge\recovered\papers_output\CQE-paper-23.md

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Status

Horizon/speculative layer with explicit obligations

Proof/Exposure Hierarchy

The proof-carrying content of this paper is the mathematics: the definitions, lemmas, constructions, examples, and receipts that establish the claimed transport. Paper 00, workbook sheets, analog tools, and open-obligation ledgers are supplemental validation and exposure layers. They exist to make the math inspectable, reproducible, and accessible without requiring a particular software stack. In the simplest case, the same state transitions can be marked with ordinary physical tokens, lines, or dirt; the point is not the material, but the preserved center, boundary, transform, residue, and receipt structure.

Abstract

This paper states a local transport problem, gives a formal vocabulary for it, binds it to a ForgeFactory/Rhenium tool surface, and records an analog workbook sheet as supplemental exposure and validation evidence. The paper is written as a proof-facing document rather than as a description of how the paper was produced. Build-method details are retained only in appendices, receipts, and the Paper 31 meta-walkthrough.

Central Thesis

Apply the same ribbon/bifurcation logic to protein-chain fold hypotheses with contact-map and topology receipts.

Scope Boundary

The paper claims only what its math, proof tree, citations, tool receipts, and supplemental workbook evidence can presently support. Any interpretation that exceeds that support is logged as an obligation rather than silently promoted to proof.

Definitions

- **C**: the active center of a readout window.
- **L/R**: the two opposed boundary directions read relative to C.
- **Transport row**: a typed record that carries a claim, source, transform, state, and obligation status.
- **Receipt**: a replayable record of an operation, its inputs, outputs, and unresolved obligations.
- **Supplemental workbook sheet**: a supplemental physical version of the proof state, expressed through color, tokens, string, page, card, and white/black follow-up.
- **Tool binding**: the ForgeFactory module or function family that makes the paper executable or testable.

Axioms

Axiom 23.1 - Locality: every accepted claim must be readable through a local window before it is lifted to a larger frame.

Axiom 23.2 - Receipt Preservation: no transform is accepted unless its inputs, output, and unresolved residue can be logged.

Axiom 23.3 - Boundary Positivity: failed, partial, or mismatched routes are data. They become obligations or correction surfaces.

Axiom 23.4 - Analog Exposure Equivalence: if the claim is part of the main corpus, it can be exposed through a physical workbook analogue.

Lemmas

Lemma 23.1 - If a local state preserves C and records L/R residue, it can be transported into a proof ledger without erasing unresolved alternatives.

Lemma 23.2 - If a tool output and supplemental workbook sheet encode the same center, boundary, and obligation state, they are equivalent receipts at different media layers.

Lemma 23.3 - A practical example is valid for this paper only when it demonstrates the same operation in a non-toy domain.

Formalism / Calculus Sketch

Let a paper state be P = (C, L, R, B, T, O), where C is the center, L and R are boundary readouts, B is the boundary rule, T is the tool transform, and O is the obligation set.

A local transform is accepted when:

```
T(P_in) -> P_out
receipt(P_in, T, P_out, O) exists
C_out is defined
unresolved residue is in O rather than erased
```

For Paper 23, the tool binding is:

```
forgefactory.foldforge
```

Proof Tree

```
claim
-> local window
-> boundary read
-> tool transform
-> practical example
-> supplemental workbook analogue
-> receipt
-> proof result / audit residue split
```

Practical Solved Example

Domain: short peptide chain mapped into fold candidates and untested contact obligations.

Procedure: define a center, identify left/right or equivalent boundary states, run or simulate the tool transform, record any failed or incomplete path as an obligation, and export a receipt.

Solved Output: the example is solved when the operator can reproduce the same state transition from the formal paper, the ForgeFactory tool, and the analog workbook sheet.

Tool Binding

- Module: `forgefactory.foldforge`
- Required outputs: `receipt.json`, `workbook_sheet.json`, `example_result.json`, `obligation_ledger.csv`.
- Minimum test: a smoke test that produces at least one proof-like row and one obligation-like row.

Analog Workbook Sheet

- Start with grey loose substrate.
- Place C token at the local center.
- Mark active color gradients: red, green, blue minimum.
- Use string to bind the main route.
- Use white follow-up for proof continuation.
- Use black follow-up for obligation or unresolved residue.
- Bind the final sheet into the matching color notebook.

IRL Citation Anchors

- [AlphaFoldNature] Jumper et al. Highly accurate protein structure prediction with AlphaFold. Nature 2021. URL: https://www.nature.com/articles/s41586-021-03819-2 Use: protein folding / structure prediction anchor.
- [ProteinKnots] Brems et al. AlphaFold predicts the most complex protein knot and composite protein knots. URL: https://arxiv.org/abs/2207.07410 Use: topological complexity and knots in protein backbones.

Open Obligations

- Replace citation anchors with final bibliography entries during peer-review preparation.
- Expand the tool binding from stub/registry level to executable tests where not yet implemented.
- Add one domain expert critique pass.
- Add one falsifier case that the tool must reject or defer.

Back-Propagation Targets

- Paper 00 receives any new contract term needed here.
- ForgeFactory registry receives or updates `forgefactory.foldforge`.
- The analog workbook manual receives any new sheet rule required by this paper.
- Paper 31 records how this paper's presentation order demonstrates the same LCR process.
